

Lecture 4

Shape and Graphs of Distributions

Review

1 What is a distribution?

The values a variable takes and how often each one occurs.

2 What are the three features we look for in a distribution?

Shape, center, and variability.

3 Which feature do we describe for a categorical variable?

The modal category – the category with the highest frequency.

4 What is a frequency table?

A table of the distinct values of a variable together with the number of observations of each value.

5 Why can't we build a frequency table for a continuous variable directly?

There are too many distinct values – almost every observation is unique. We bin first.

Review

1 How do we bin a continuous variable?

Divide the range into K non-overlapping intervals of equal width, then count how many fall in each.

2 Which graphs do we use for a qualitative variable?

Bar graphs, pie charts, and Pareto charts.

3 Which graphs do we use for a quantitative variable?

Dot plots, stem and leaf plots, and histograms – today's topic.

4 Why graph the data when we already have the frequency table?

The graph shows shape, center, and spread at a glance, along with any outliers.

Shape of a distribution

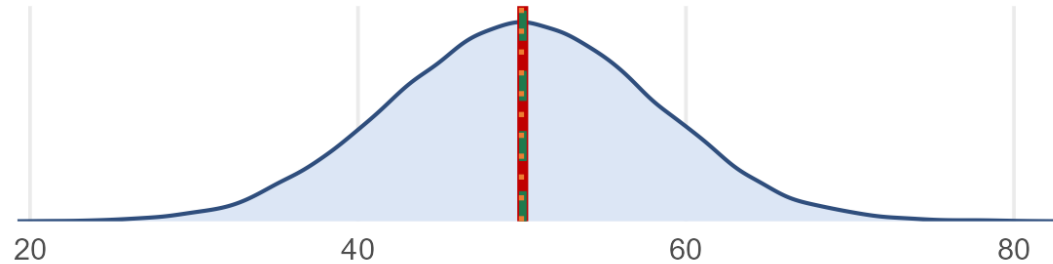
Mean – the balancing point of the data. Add up every observation and divide by how many there are.

Median – the middle value. Half of the observations fall below it and half fall above.

Mode – the most common value, which is the peak of the distribution.

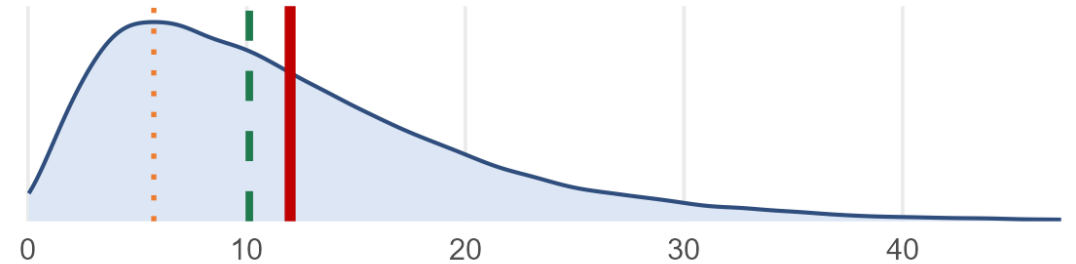
Symmetric

mean = median = mode: all three land in the same place



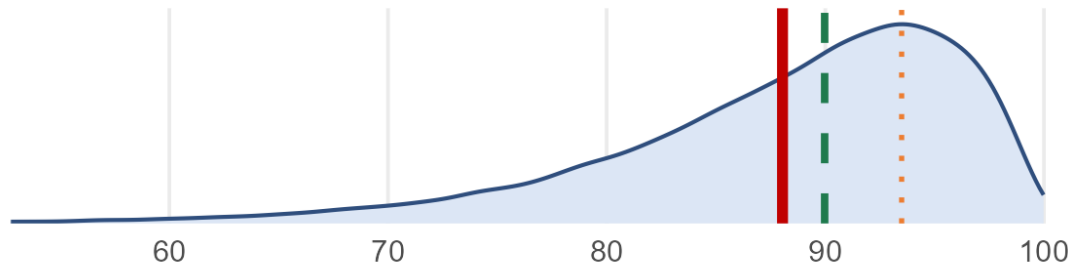
Skewed right

mode < median < mean: the tail pulls the mean right



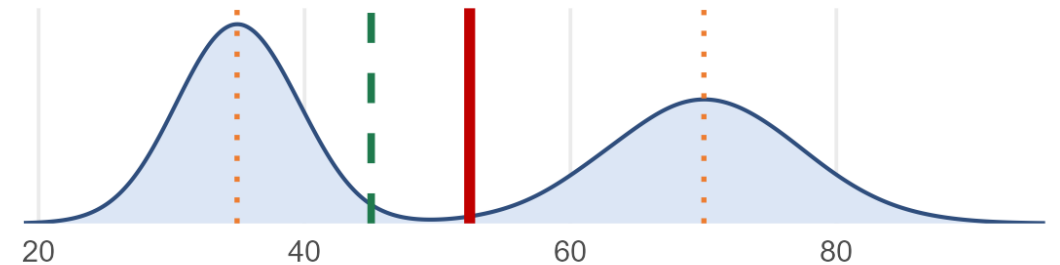
Skewed left

mean < median < mode: the tail pulls the mean left



Bimodal

two modes, and the mean and median fall in the gap between them



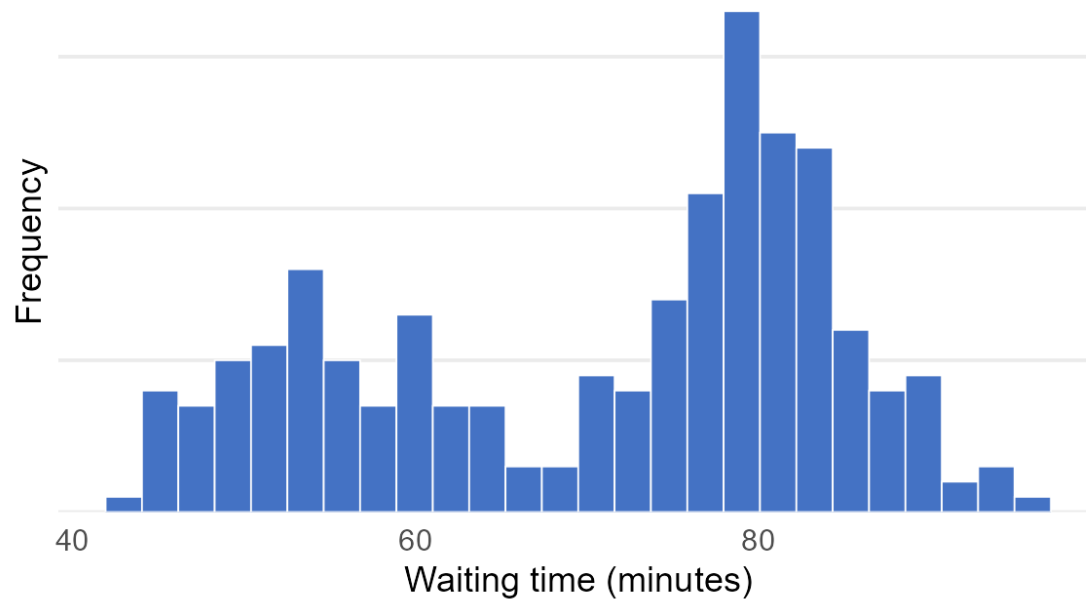
--- Mode - - - Median - - - Mean

Bimodality: Old Faithful

Two peaks are a signal that the sample mixes two different kinds of thing. Here, two kinds of eruption.

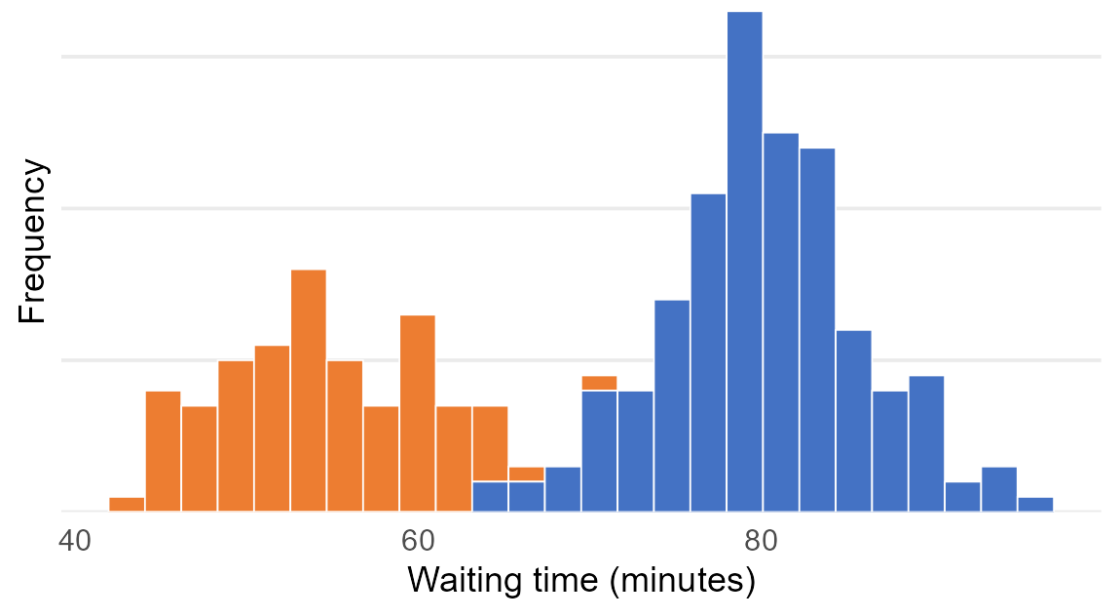
Old Faithful: waiting time between eruptions

272 eruptions. Two clear humps - about 55 minutes, and about 80



The same waits, split by how long the eruption lasted

Each group is a single hump. The gap was two kinds of eruption



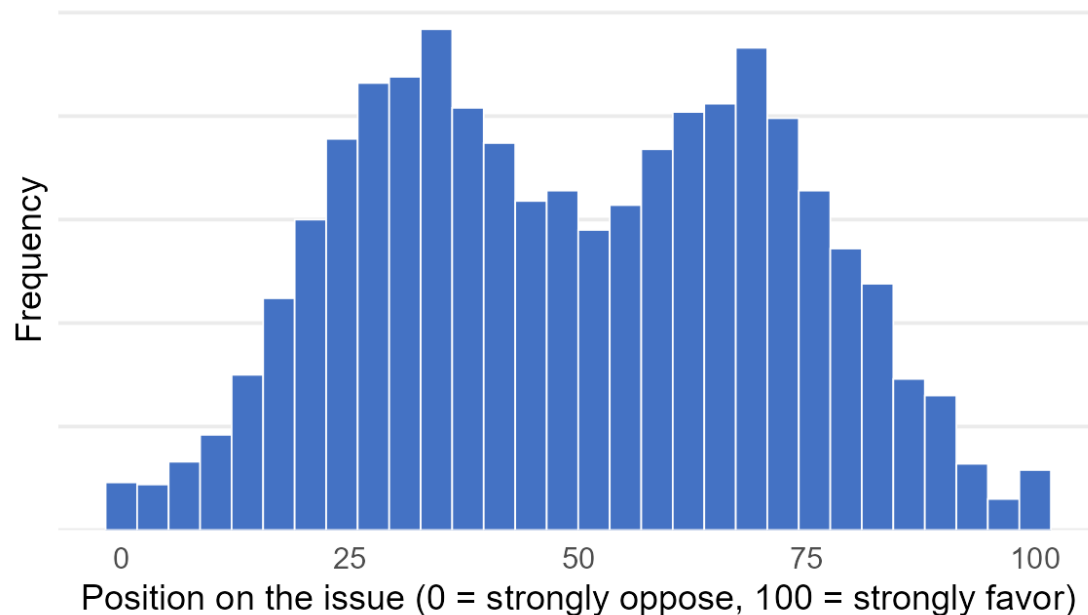
Short eruption (< 3 min) Long eruption (3 min or more)

Bimodality: a polarized public

A population polarized on an issue IS two sub-populations. Here the grouping variable is party.

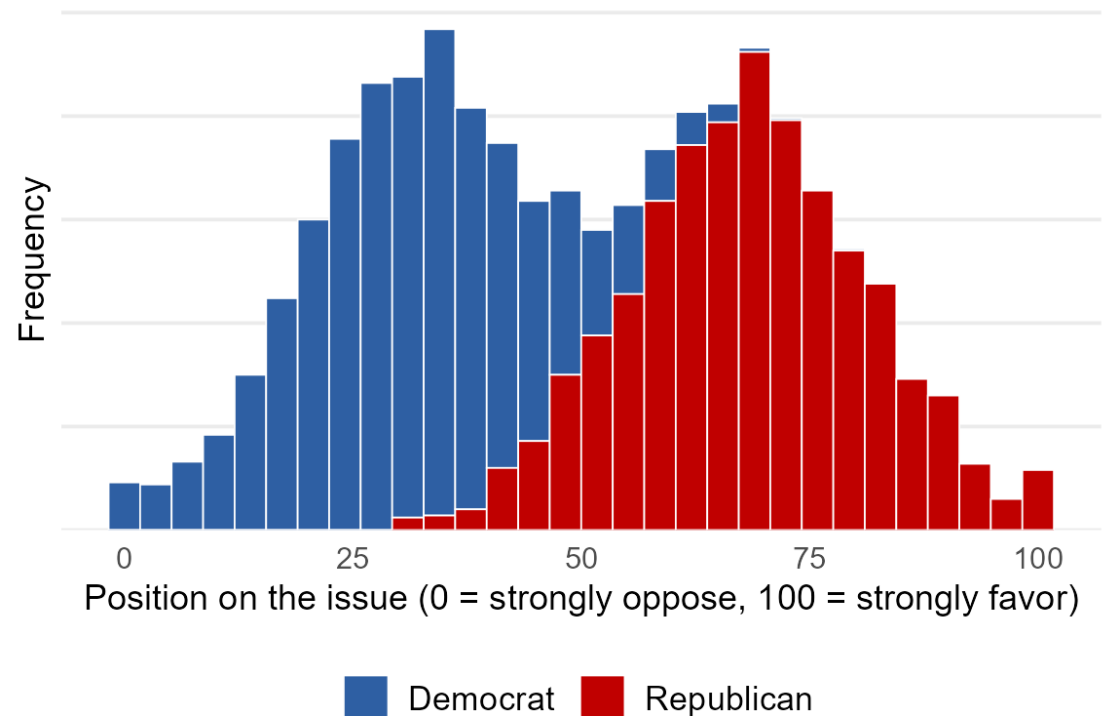
Attitudes on a contested political issue

Simulated for illustration. The middle is thinner than either side



The same responses, split by party

Polarization IS population structure: two groups, each one hump

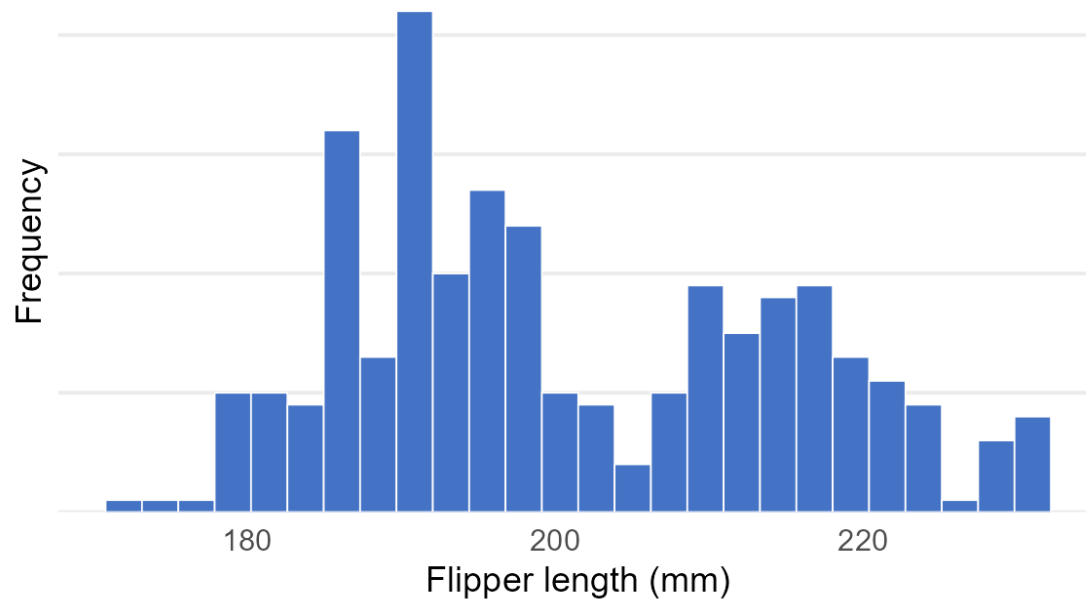


Bimodality: penguin species

Here the grouping variable was recorded, so we can check the claim instead of just arguing it.

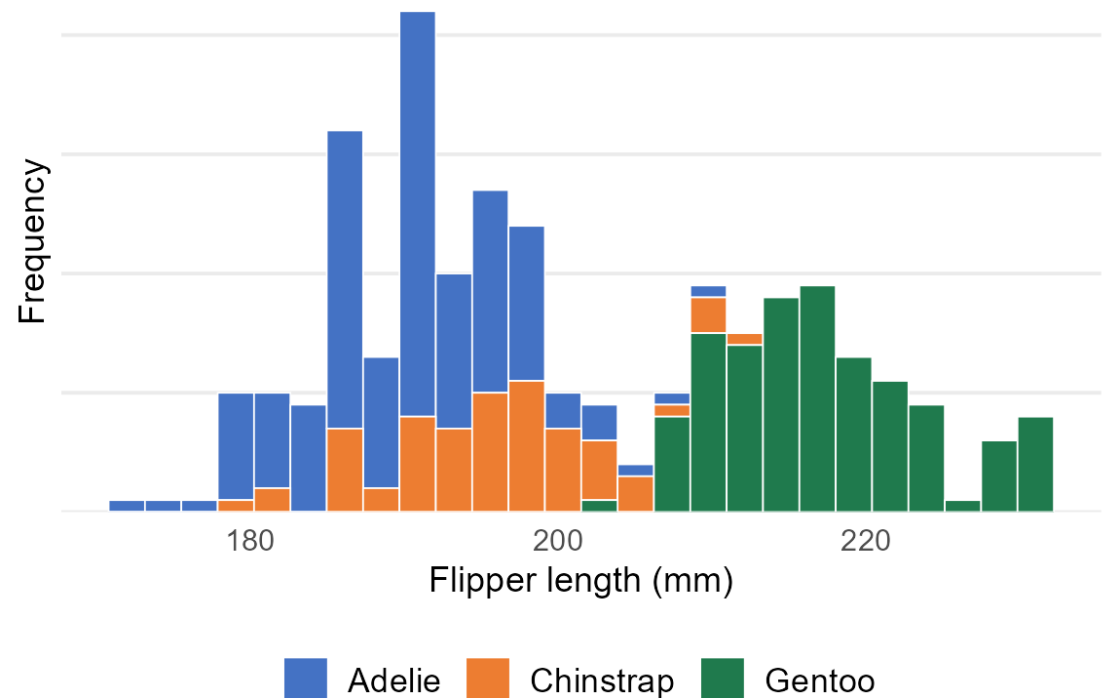
Penguin flipper length, Palmer Archipelago

342 penguins. Two humps, and nothing in particular at 205 mm



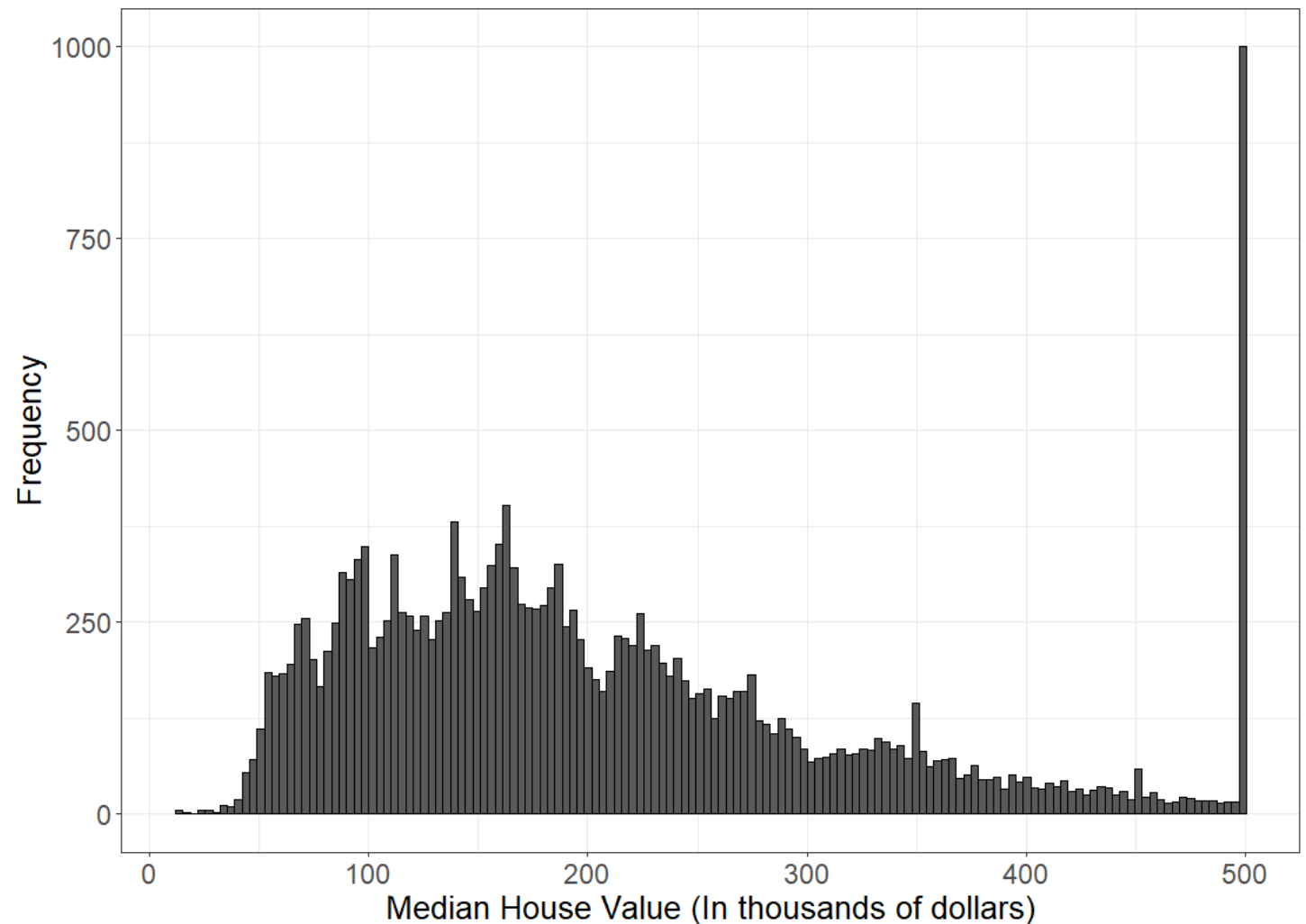
The same flippers, split by species

Gentoo are simply bigger birds. The gap was three species all along



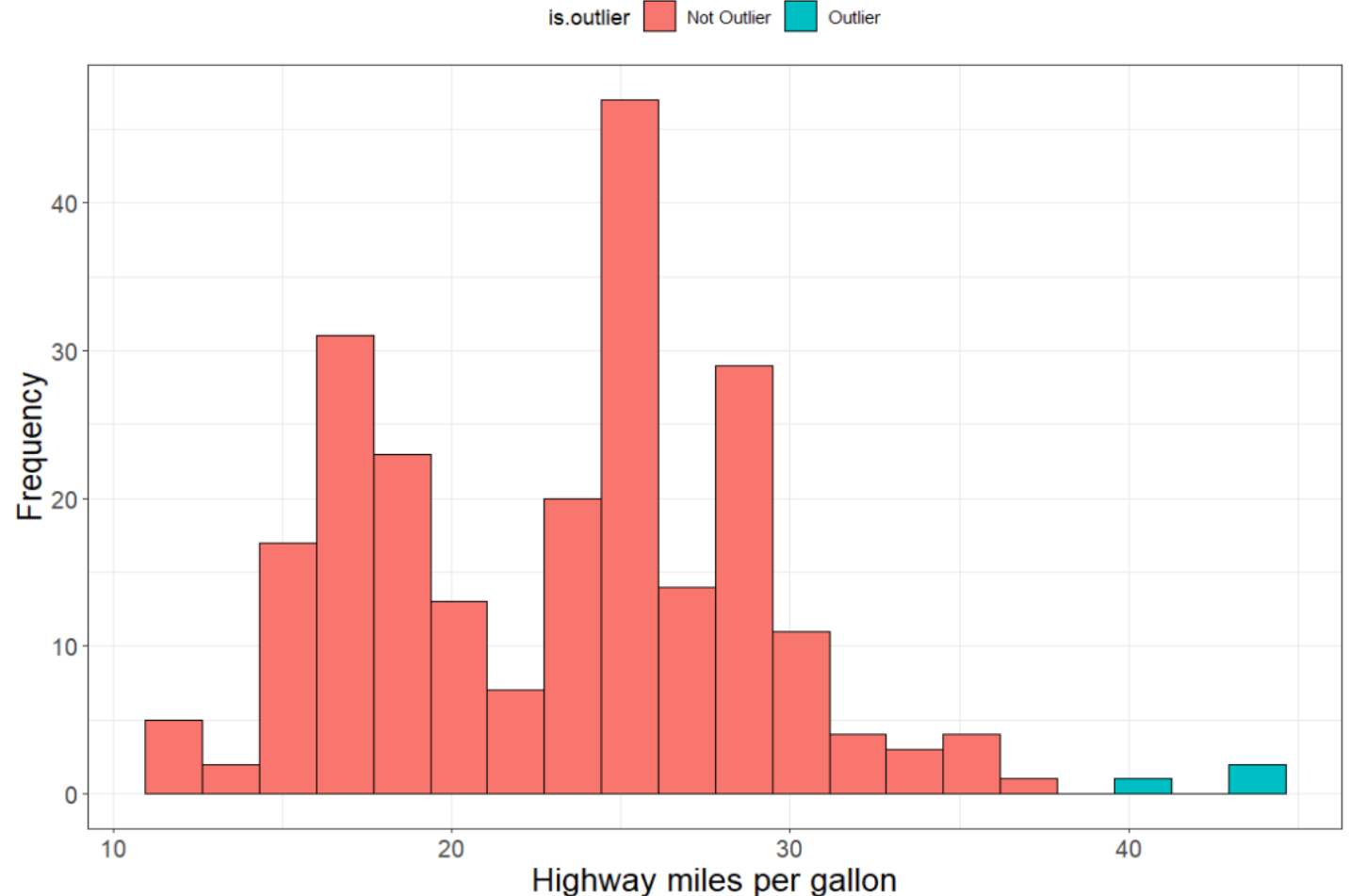
Skew: a hard limit on the values

- Skew appears when there is a hard limit on the values a variable can take
- Values pile up against the limit and trail away from it
 - Prices and incomes cannot go below zero
 - The age of adults cannot go below 18
 - Exam scores cannot go above 100
- The long tail drags the mean toward it, so the mean and median disagree



Outliers: a real value, or a mistake

- An outlier is an extreme value lying far away from the rest of the data
- Two very different causes:
 - a real observation that genuinely is unusual
 - a measurement or recording error
- The plot cannot tell you which. Go back to the source and check
- Either way an outlier pulls the mean hard, so find them first

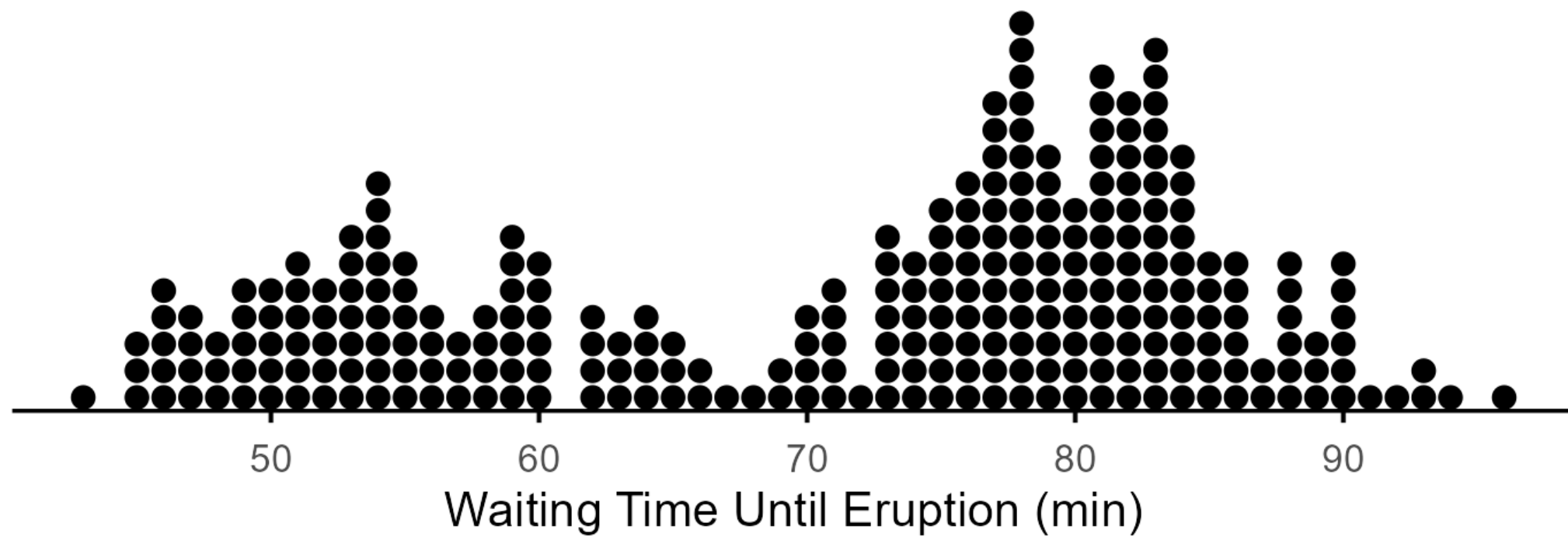


Visualizing Distributions: Quantitative Variables

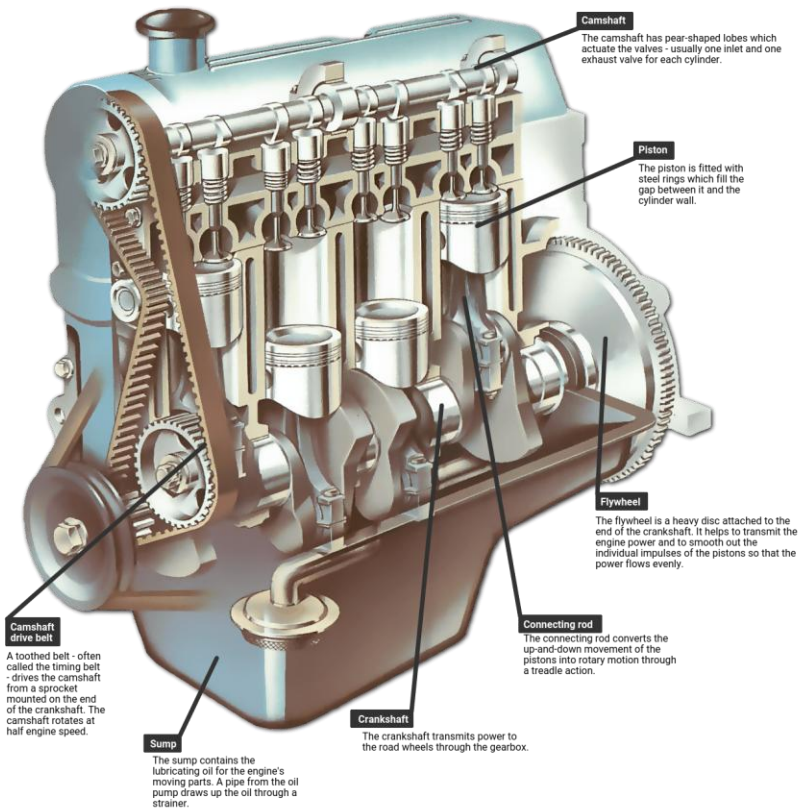
- **Dot plots** – shows a dot for each observation placed above the value for that observation

Steps to construct a dot plot

1. **Draw a horizontal line and mark the line with regular values of the variable**
2. **For each observation, place a dot above its value on the number line**
 - Works best with quantitative discrete data
 - Doesn't work well if the variable is continuous and takes on many distinct values...
 - For continuous data, the values may need to be round to the nearest tenth or integer



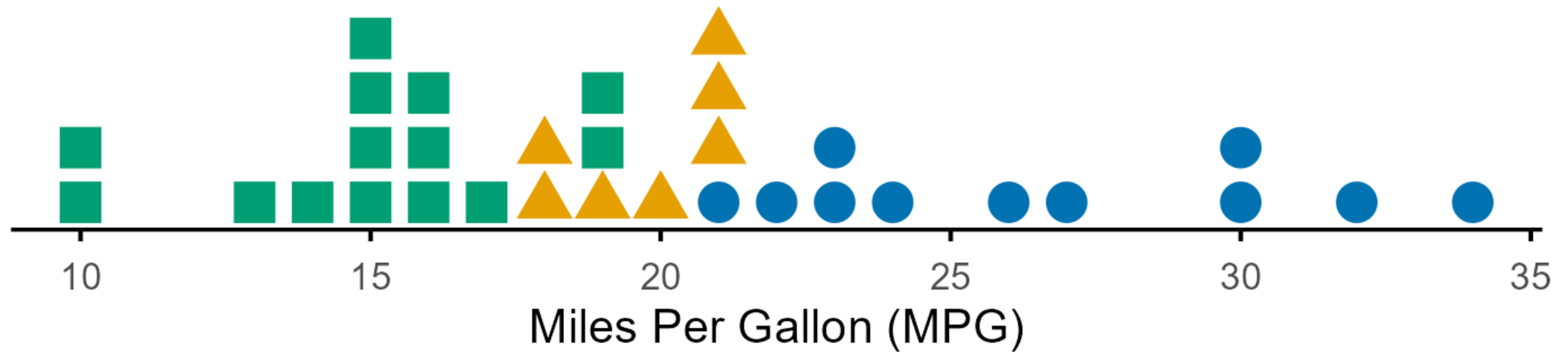
Example: MPG and Engine Cylinders



Observation	MPG	Cylinders	Model
1	21.0	6	Mazda RX4
2	21.0	6	Mazda RX4 Wag
3	22.8	4	Datsun 710
4	21.4	6	Hornet 4 Drive
5	18.7	8	Hornet Sportabout
6	18.1	6	Valiant
⋮	⋮	⋮	⋮
32	21.4	4	Volvo 142E

About the data: mtcars – road-test figures for 32 cars from the 1974 Motor Trend US magazine (1973–74 models), one row per car. We use two of its 11 variables: mpg (miles per US gallon, ranging 10.4 to 33.9) and cyl, the number of engine cylinders – 4, 6 or 8, with 11, 7 and 14 cars in each group. Cylinder count is a grouping variable, which is what lets the dot plot on the next slide compare three groups side by side.

Number of Cylinders ● 4 ▲ 6 ■ 8



Dot plot: strengths and weaknesses

- **Strengths**

- Shows every individual observation, so nothing is hidden or summarized away
- Quick to draw by hand, and easy for a reader to check against the data
- Good for small samples and for comparing a few groups side by side

- **Weaknesses**

- Becomes an unreadable smear once n is large or values repeat heavily
- Needs rounding for continuous data, which throws away precision
- Cannot be smoothed the way a histogram can with wider bins

Visualizing Distributions: Quantitative Variables

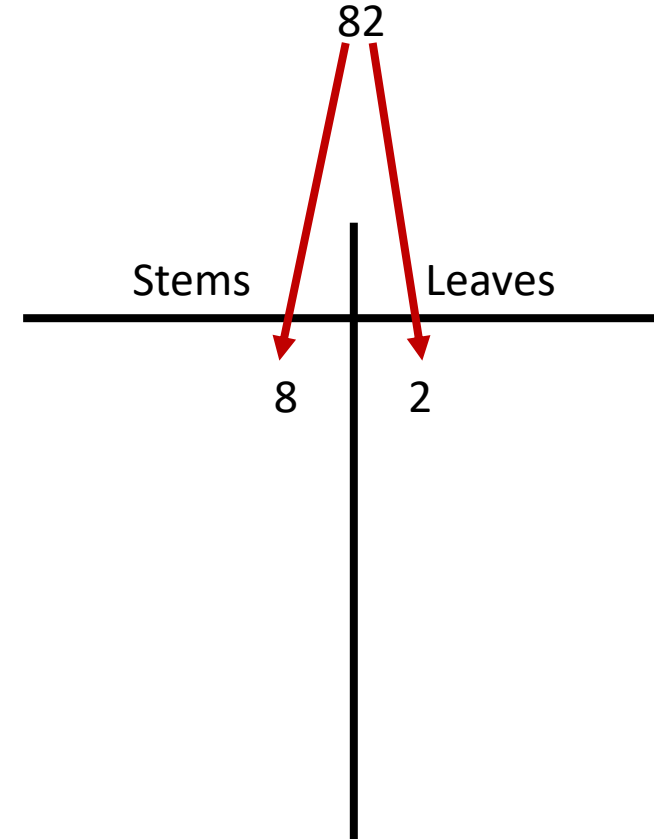
Stem and leaf plot – like a dot plot, a stem and leaf diagram also displays individual observations.

Stem – all the digits in an observation except the last digit

Leaf – the last digit in an observation

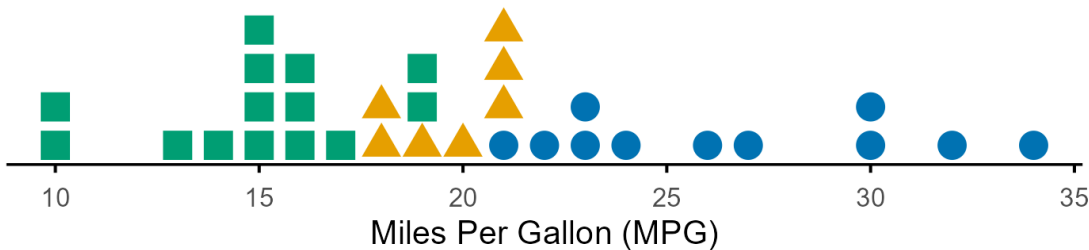
Steps to construct a stem and leaf plot

1. Sort the data in order from smallest to largest.
2. Place the stems in a column in increasing order
3. Place a vertical line to the right of the stems
4. To the right of the vertical line, fill in the leaves that correspond with each stem in increasing order



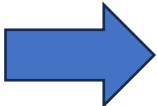
Example: MPG

Number of Cylinders ● 4 ▲ 6 ■ 8



Observation	MPG	Cylinders	Model
1	21.0	6	Mazda RX4
2	21.0	6	Mazda RX4 Wag
3	22.8	4	Datsun 710
4	21.4	6	Hornet 4 Drive
5	18.7	8	Hornet Sportabout
6	18.1	6	Valiant
⋮	⋮	⋮	⋮
32	21.4	4	Volvo 142E

Stems	Leaves
10	4,4
11	
12	
13	3
14	3,7
15	0,2,2,5,8
16	4
17	3,8
18	1,7
19	2,2,7
20	
21	0,0,4,4,5
22	8,8
23	
24	4
25	
26	0
27	3
28	
29	
30	4,4
31	
32	4
33	9



Stems	Leaves
10	4,4
12	3
14	3,7,0,2,2,5,8
16	4,3,8
18	1,7,2,2,7
20	0,0,4,4,5
22	8,8
24	4
26	0,3
28	
30	4,4
32	4,9

Try it out: Stem and leaf plot

Data = 4.2, 3.8, 4.6, 3.2, 2.7, 8.2, 9.1, 0.2, 1.2, 6.2

Stem and leaf plot: strengths and weaknesses

- **Strengths**

- Keeps the actual data values, so the original observations can be read back off the plot
- Shows the shape and the digits at the same time
- Works well for moderate samples, roughly n of 15 to 50

- **Weaknesses**

- Awkward for large n : the leaves run off the page
- A poor fit for decimals, or for values spanning many orders of magnitude
- The shape depends on which digit you choose as the stem

Visualizing Distributions: Quantitative Variables

Stem and leaf plots and **dot plots** are unwieldy for large n

Histogram – uses bars to portray the frequencies or relative frequencies of the possible outcomes for a quantitative variable

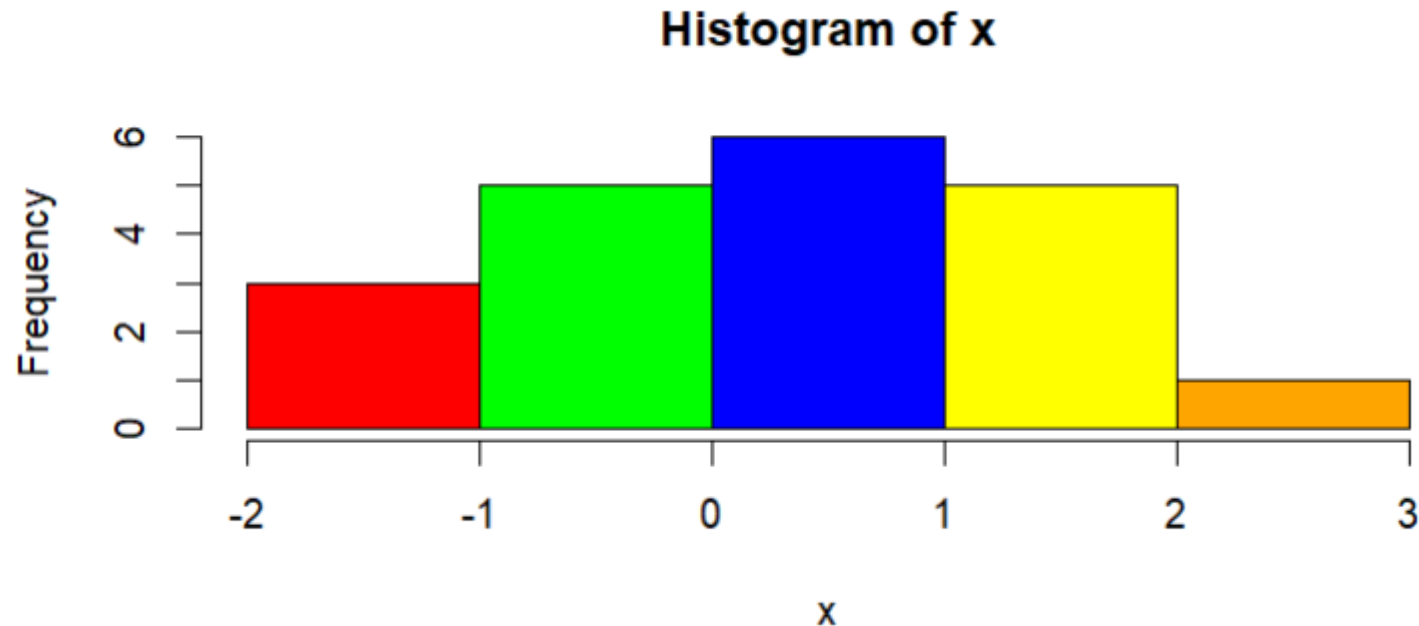
Steps to construct a histogram

- 1. Divide the range of the data into intervals of equal width**
- 2. Compute the frequency of each interval (i.e construct the frequency table)**
- 3. Label the x-axis with the values or endpoints of each interval.**
- 4. Draw a bar over each value or interval with height equal to its frequency or relative frequency**

Try it out: Histogram

Consider the following $n = 20$ observations of a continuous variable

Bin 1	Bin 2	Bin 3	Bin 4	Bin 5
-1.5, -1.2, -1.0,	-0.8, -0.7, -0.6, -0.1, -0.1,	0.1, 0.1, 0.1, 0.6, 0.6, 0.8,	1.1, 1.2, 1.3, 1.8, 1.9,	2.4



How to choose the number of Bins?

- How to choose the best number of bins is not a straightforward question and there is a lot of literature on the subject
- We can construct our histogram using a specific binwidth w or under a set number of bins k
- $w = \frac{\max x - \min x}{k}$ or $k = \frac{\max x - \min x}{w}$
- **Square root method:** $k = \text{round}(\sqrt{n})$ (A fairly safe and basic rule of thumb)
- Sturges Rule^[1]: $k = \text{round}(\log_2 n) + 1$ (not great for $n < 30$)
- Rices Rule^[2]: $k = 2\sqrt[3]{n}$

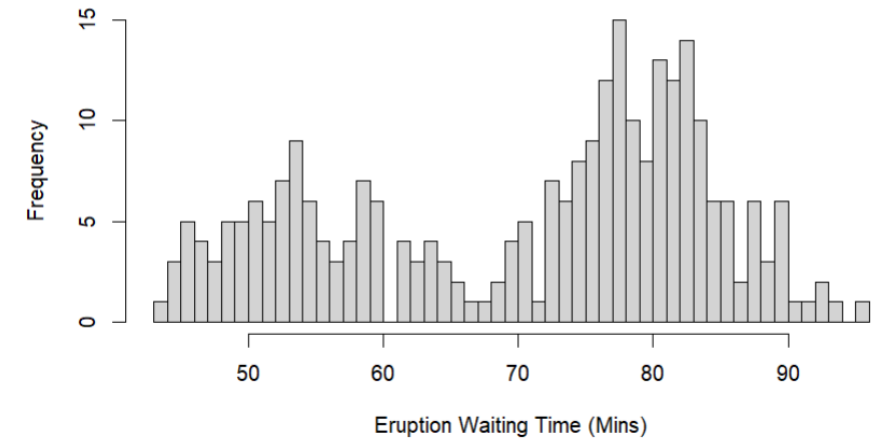
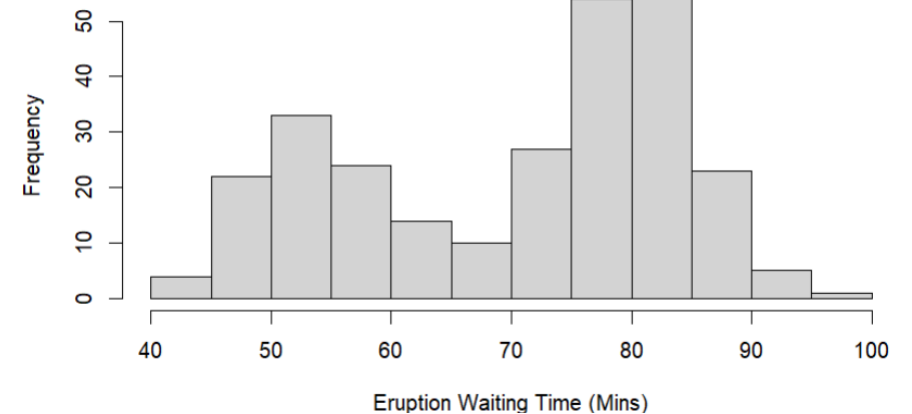
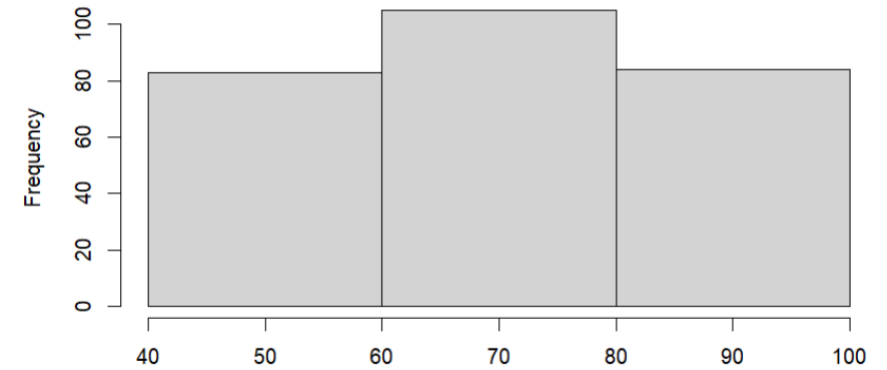
[1] Sturges, Herbert A. "The choice of a class interval." Journal of the american statistical association 21.153 (1926): 65-66.

[2] Lane, David. *Online statistics education: A multimedia course of study*. Association for the Advancement of Computing in Education (AACE), 2003. – Chapter 2 “Graphing Distributions

Some tips

- If too few intervals are used, then the graph will be too crude
- If too many intervals are used, graph will contain many short bars and gaps.
Usually between 5 - 15 intervals are enough.
- Most plotting software will automatically choose the number of bins.
- **ALWAYS** plot the histogram to get an idea about the shape of the distribution of a quantitative variable
- Is the number of observations is small (say $n < 50$) then it's a good idea to supplement a histogram with a dot plot or stem plot

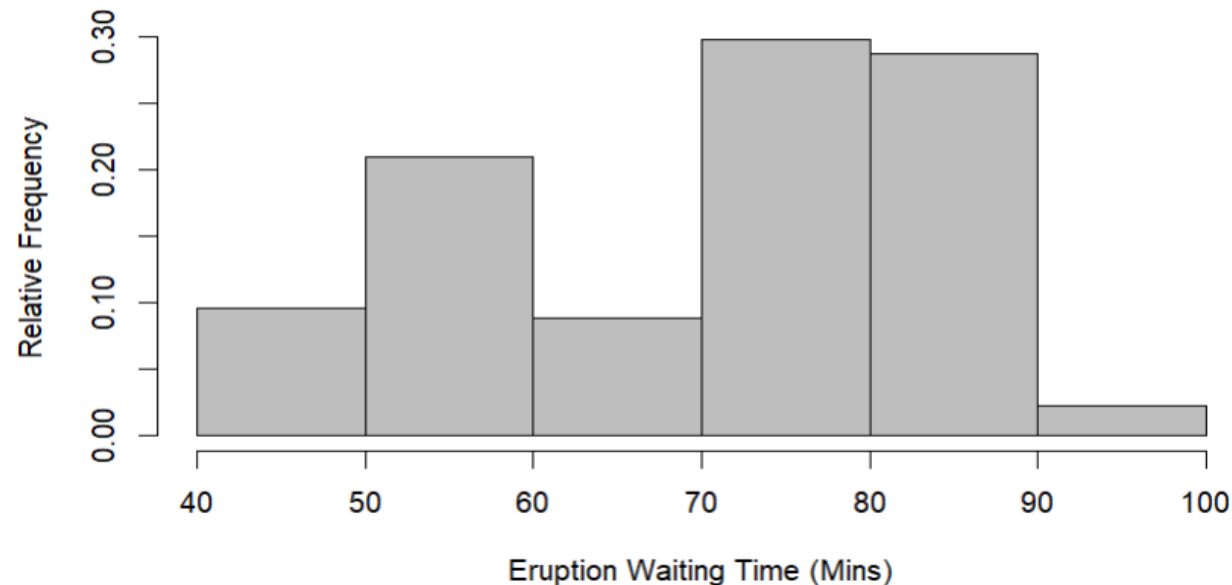
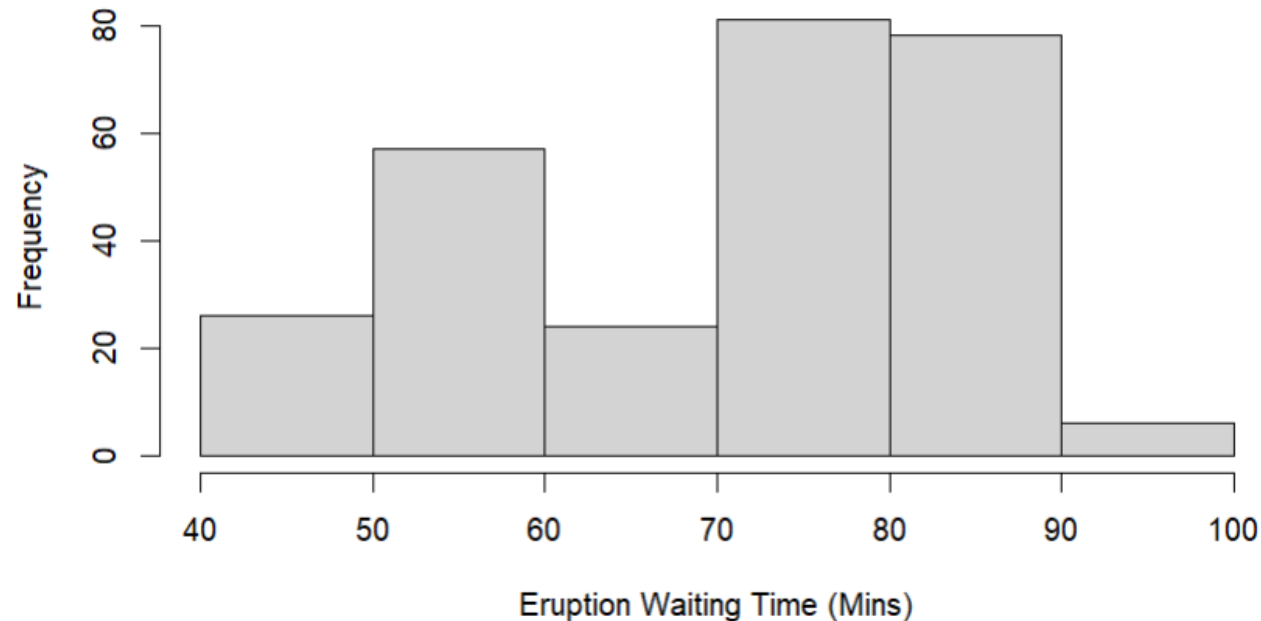
Histogram of Eruption Waiting Times



Example: Old Faithful Eruption Times

Waiting Time (Min)	Frequency	Relative Frequency	Cumulative Relative Frequency
< 50	21	0.077	0.077
50 - 60	56	0.206	0.283
60 - 70	26	0.096	0.379
70 - 80	77	0.283	0.662
80 - 90	80	0.294	0.956
> 90	12	0.044	1

Histogram of Eruption Waiting Times



Histogram: strengths and weaknesses

- **Strengths**

- A great way to get an idea of shape when the sample is medium or large ($n > 20$)
- Handles continuous data at any sample size, and stays readable as n grows
- The standard choice for showing a distribution to someone else

- **Weaknesses**

- Hides the individual observations: you see bars, not data points
- The shape depends on the number of bins, so it can distort shape for small samples
- Bin edges are arbitrary, and moving them can change the story the picture tells

From pictures to numbers

- **Next: measures of location and position.** Single numbers that say where a distribution sits.
- **Numbers can be computed with.** A graph is read by eye. A statistic goes into a table, a comparison, or a hypothesis test.
- **"What is a typical value?"** The first question anyone asks of a dataset, and the mean, median and mode each answer it differently.
- **Comparing groups.** Two samples are compared through their centers, not by squinting at two histograms.
- **Where does one observation sit?** A score of 78 means little until you know it falls at the 90th percentile.
- **Shape decides which measure to trust.** Skew and outliers drag the mean but leave the median alone, which is why we did shape first.

Measures of Central Tendency

- The (arithmetic) **mean** is the average value of a set of observations
it measures the center of mass of a distribution (the balancing point)

$$\bar{x} = \frac{x_1 + x_2 + \dots + x_n}{n} = \sum_{i=1}^n \frac{x_i}{n}$$

- the mean is usually not equal to any of the values observed in the sample
- The mean is highly influenced by **outliers** - observations that take on extreme values relative to the distribution

Practice: Calculate The Mean

- $X = \{1, 3, 5, 5, 6, 7, 7, 8\}$

Measures of Central Tendency

- The **median** is the middle value of a set of observations

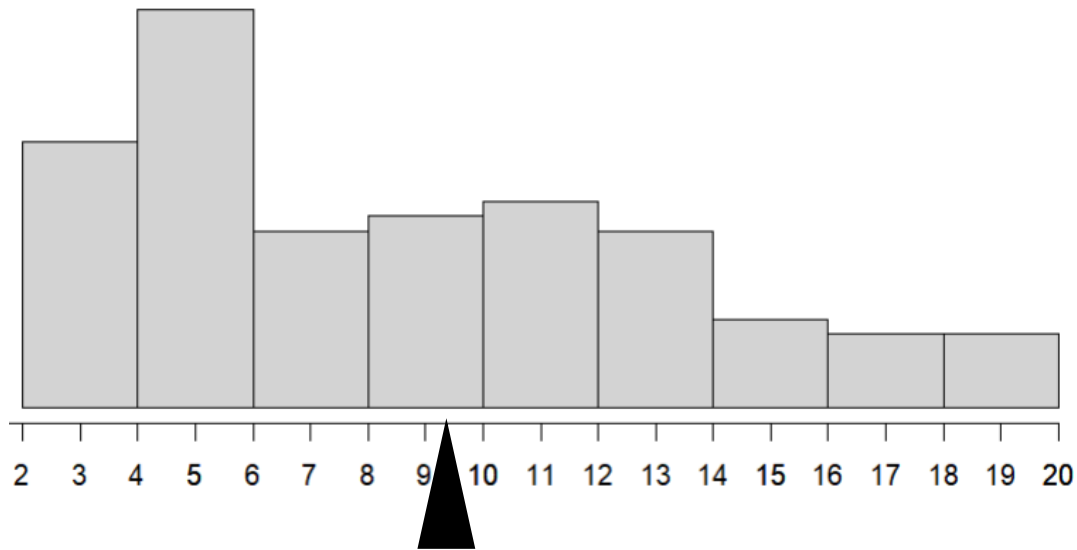
How to compute the median:

1. Compute the median by first ordering the observations from smallest value to largest value and choose the number in the middle
2. If the n is odd the median is the middle number
 - If n is even the median is the sum of the two middle values divided by 2

Practice: Calculate the Median

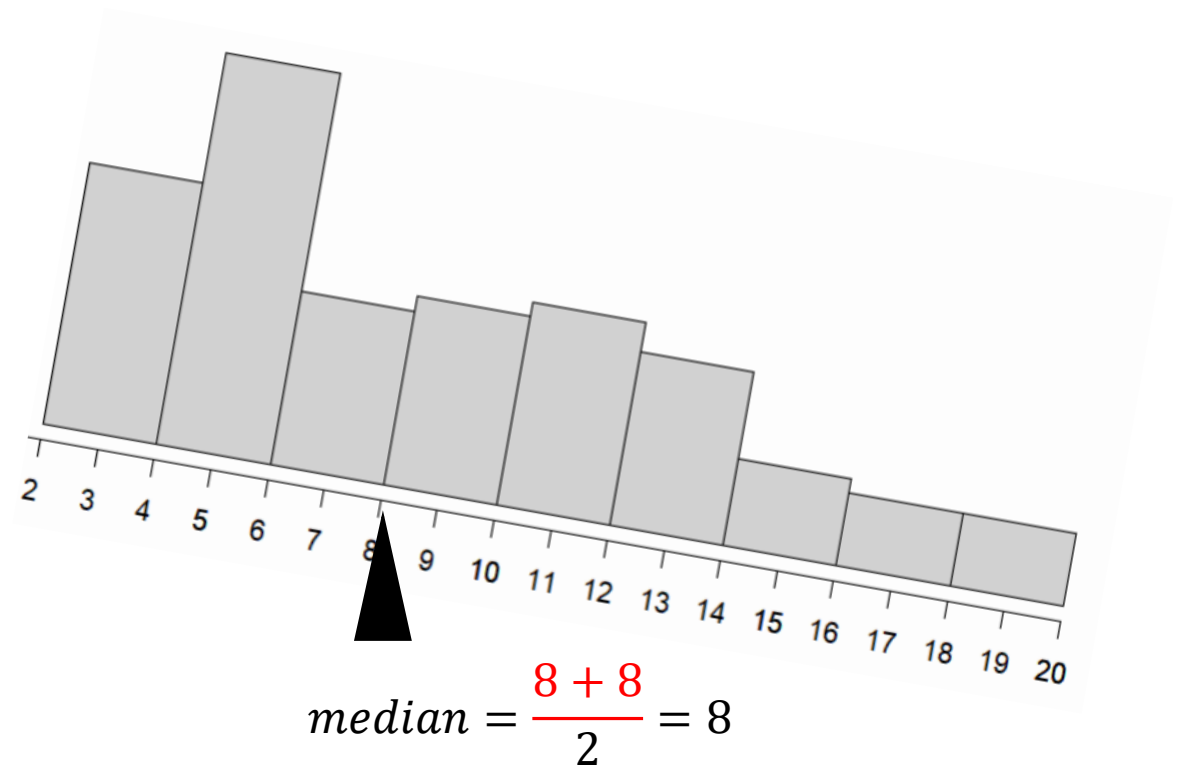
- $X = \{1, 3, 5, 5, 6, 7, 7, 8\}$

Data: 3 3 3 3 3 3 3 4 4 4 4 4 4 4 4 4 5 5 5 5 5 5 5 5 5 5
 5 5 5 5 5 5 5 5 5 5 6 6 6 6 6 6 6 6 6 6 7 7 7 7 7 7 7
 7 8 8 8 8 8 9 9 9 9 9 9 9 10 10 10 10 10 10 10 11 11 11 11 11 12
 12 12 12 12 12 12 12 13 13 13 14 14 14 14 14 14 15 15 15 16 16 16 16 17 17 17 18
 18 20 20 20 20 20



$$\bar{x} = 9.2$$

The mean is the center of gravity

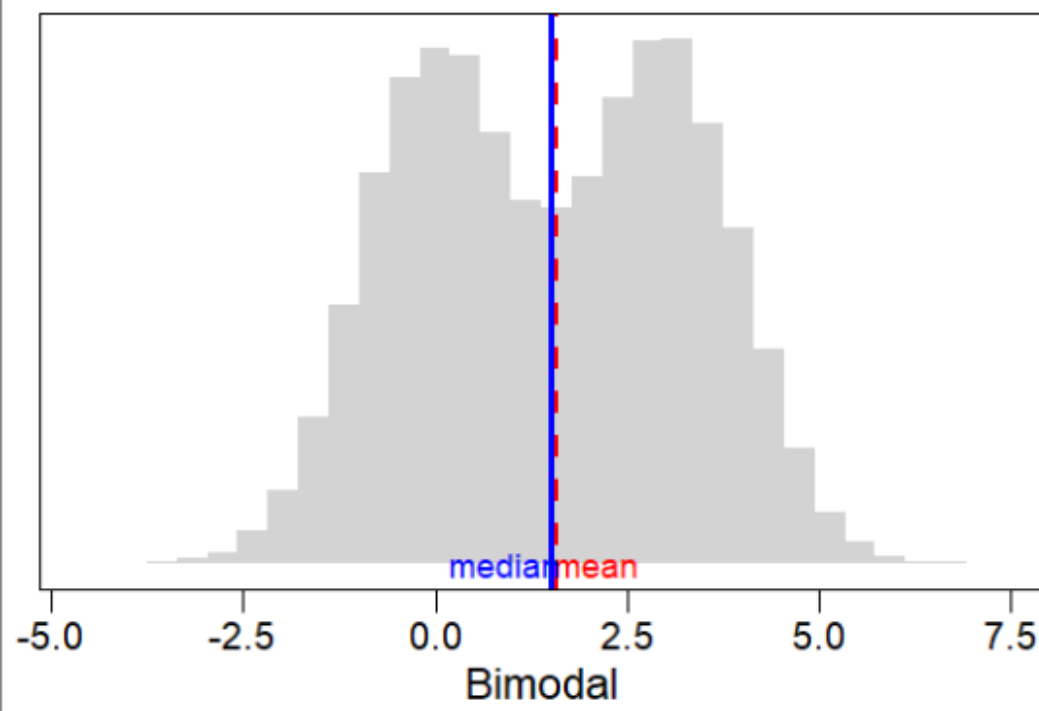
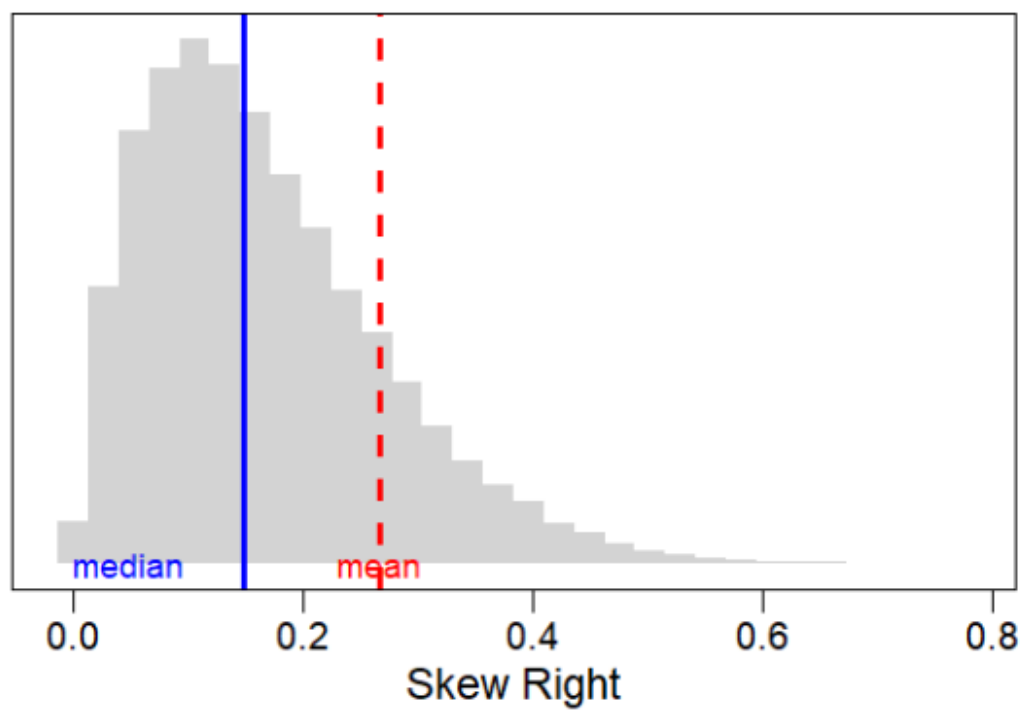
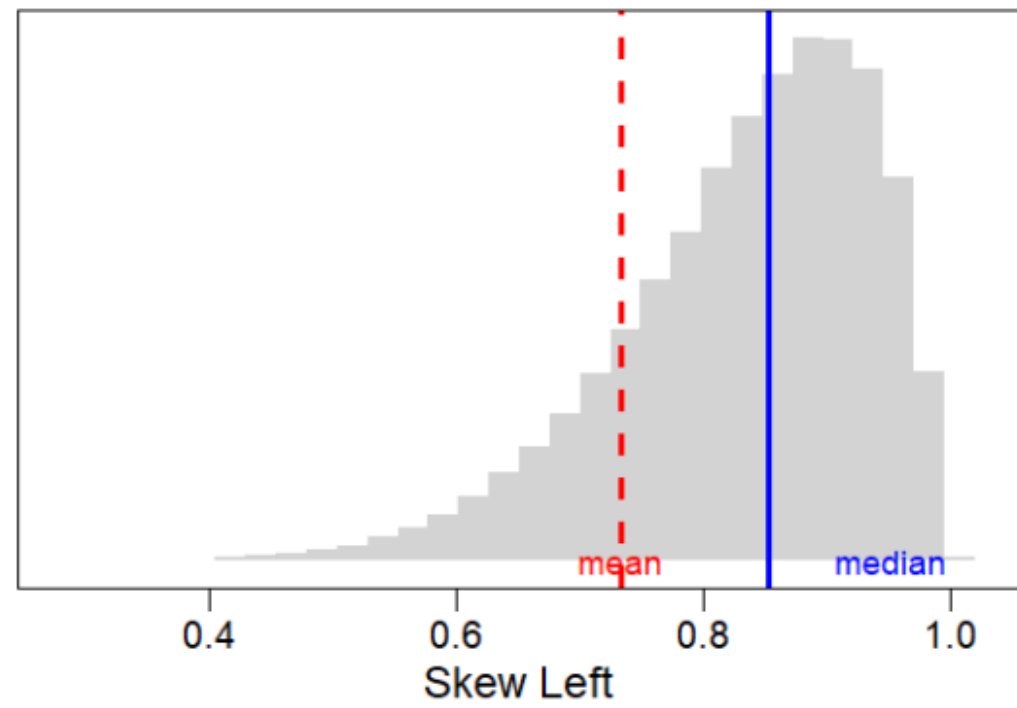
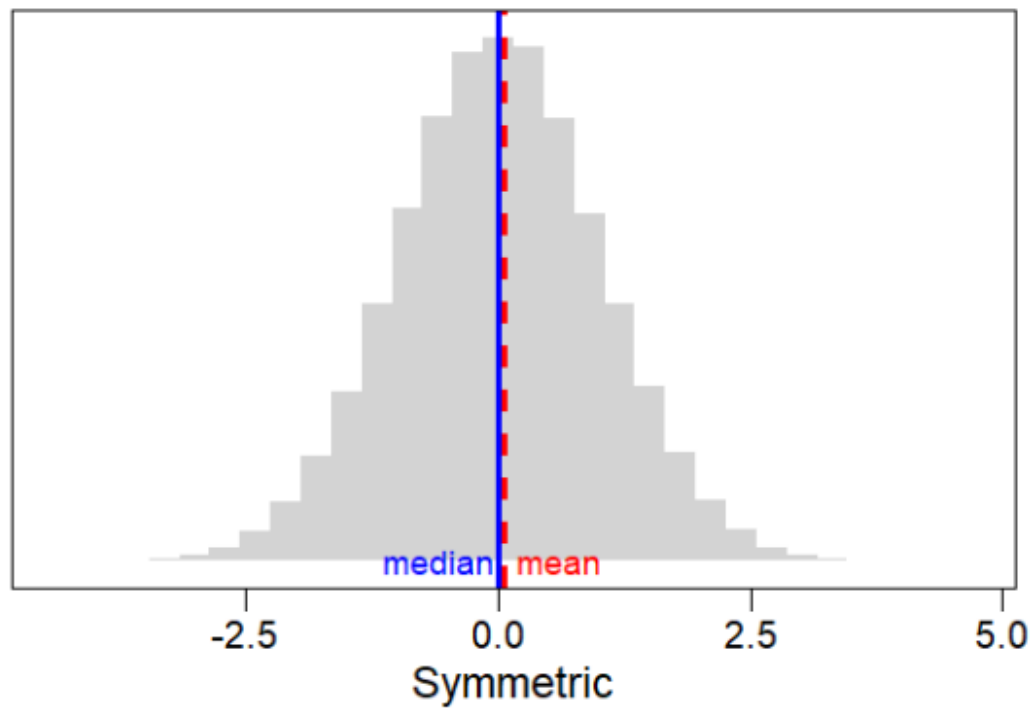


$$median = \frac{8 + 8}{2} = 8$$

The median is the middle value

Mean and median treat outliers differently

- $X = \{1, 3, 5, 5, 6, 7, 7, 8, 32\}$



Alternative formulas for the mean

- We can also express the mean in terms of the frequency F or the relative frequency RF

$$\bar{x} = \frac{1}{n} \sum_x x F(x) \quad \text{or} \quad \bar{x} = \sum_x x RF(x)$$

Where the sum is over all distinct values of the variable x

Example: Computing the mean from a frequency table

$X = \{1, 3, 5, 5, 6, 7, 7, 8\}$

X	Freq.	Rel. Freq
1	1	0.125
3	1	0.125
5	2	0.250
6	1	0.125
7	2	0.250
8	1	0.125

The mode

- The **mode** is the value with the largest relative frequency (i.e the value that occurs most often)
 - Can be used with categorical data (mean and median cannot)
 - e.g the most frequent category
 - It may not be unique if two or more values have the same frequency
 - **Caution** for quantitative data, the mode may not anywhere near the center of the distribution.

Ex.)

Data = 1,1,4,5,6

Mode = 1

Data = 1,1,4,5,6,6

Mode 1, 6

Practice:

- Roll a six-sided die $n = 10$ times and record the number rolled each time
- Data = 1,2,3,3,4,4,4,5,6,6

x	$f(x)$	$rf(x)$
1	1	0.1
2	1	0.1
3	2	0.2
4	3	0.3
5	1	0.1
6	2	0.2

Compute the **mean** using all 3 equations:

Compute the **median**

Compute the **mode**